

ITA0612

Artificial Neural Network

Presented by

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Course code

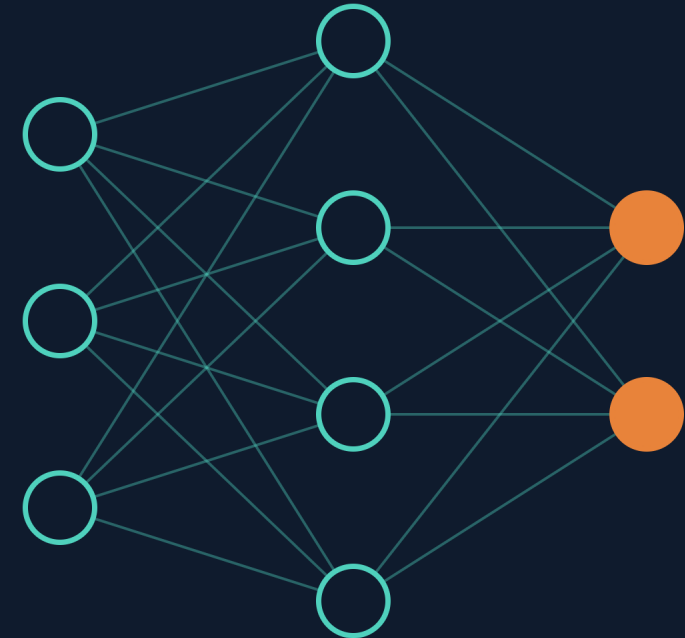
ITA0612

Register number

192312272

Faculty

Dr. Siva Kumar



What Is an ANN?

An Artificial Neural Network is a computing model inspired by the human brain. It is built from interconnected artificial neurons that learn patterns from data instead of following hand-written rules.

INSPIRED BY

The brain

Neurons passing signals through connections

MADE OF

Layers

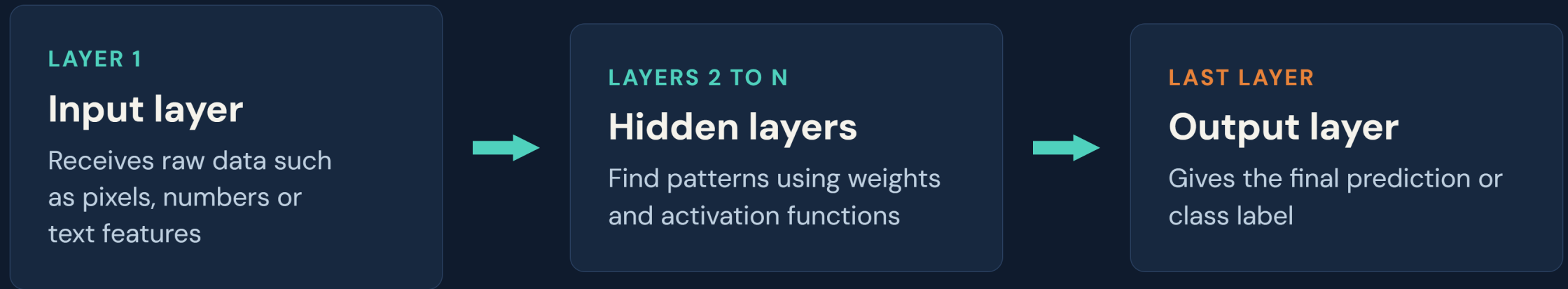
Input, hidden and output layers of connected nodes

LEARNS

From data

Adjusting connection weights to reduce errors

Architecture of an ANN



Every connection carries a weight. Learning means tuning those weights.

How a Neuron Works

$$y = f(\sum w \cdot x + b)$$

x

Inputs

Values fed into the neuron

w

Weights

Importance of each input

b

Bias

Shifts the result up or down

f

Activation

Sigmoid, ReLU or Tanh

How an ANN Learns

1

Forward pass

Data flows through the layers to give a prediction

2

Compute loss

Compare the prediction with the true answer

3

Backpropagation

Send the error backwards to find each weight's share

4

Update weights

Gradient descent nudges weights to cut the error

Repeat over many epochs until the loss is small.

Types and Applications

Network type	Typical use
Feedforward (FNN)	Prediction and classification
Convolutional (CNN)	Image and video recognition
Recurrent (RNN, LSTM)	Speech, text and time series
Generative (GAN)	Creating images and other content

Pros, Cons and Conclusion

Advantages

- Learns complex patterns from data
- Works with images, text and speech
- Improves as more data is added

Limitations

- Needs large amounts of data
- Training is computationally costly
- Hard to explain its decisions

ANNs turn data into decisions and sit behind most of modern AI.

Thank you